# Water Lost to Irrigation Scheduling

How much is at stake in the San Joaquin Valley, and how much of it Farmify can realistically recover

Prepared 28 September 2026 · Figures are order-of-magnitude estimates, with sources and assumptions stated throughout

## 1. The scale of the problem

Agriculture accounts for roughly **80% of all water consumed by human activity in California**. The San Joaquin Valley alone farms about 4.4 million acres in a typical year, producing around $20 billion in crops. It also draws down its aquifers faster than they refill: long-term groundwater overdraft runs at roughly **1.8–2.0 million acre-feet per year**. The Sustainable Groundwater Management Act requires that gap to be closed by 2040, and some groundwater agencies have already capped growers near 1.6 acre-feet per acre — against the 3–4 acre-feet a mature almond block wants.

The commonly quoted global figure is that mean irrigation efficiency is about 56%, implying some 44% of applied water is lost. That number should be handled carefully, and this summary does not lean on it:

> **A caution on the headline statistics**
>
> A November 2025 paper in *PNAS Nexus* traced the two most-cited irrigation statistics — that irrigation takes 70% of freshwater withdrawals and produces 40% of crops — through 3,500 documents and found that 60–80% of citation paths lead to sources with no supporting data. Only about 1.5% of citing documents contained actual evidence. The defensible ranges are far wider: 45–90% of withdrawals and 18–50% of production. Global "percent wasted" figures are weaker than their ubiquity suggests.

A firmer and more directly useful observation is **how irrigation decisions are actually made**. USDA survey data indicate that more than 75% of US irrigation scheduling relies on the checkbook method, crop calendars, visual inspection of the crop, or watching neighbours. In California tree nuts — among the most sophisticated growers anywhere — an ongoing survey covering nearly a million acres of almonds finds only **38–55% using crop evapotranspiration, soil moisture monitoring or plant water status**. Roughly half of a highly capitalised industry still waters on judgement.

## 2. Where the water actually goes

Water leaves a field four ways that do the crop no good: evaporation from wet soil and wetted canopy, wind drift off sprinklers, runoff to drains, and percolation down past the root zone. Hardware determines how much of each is possible; **scheduling determines how often it happens**.

The scheduling loss has one dominant cause: applying water the crop is not yet using. In Fresno, reference crop water demand swings from about 0.03 inches a day in January to 0.26 inches a day in July — close to an eight-fold range within a single season. A set length chosen so the block survives the hottest week in July, and then left alone, over-waters for most of the months either side of it. The surplus does not stay in the root zone; it drains below it.

## 3. What Farmify changes

- **It computes demand instead of assuming it.** Daily crop water use comes from the FAO-56 Penman-Monteith equation applied to the Weather Underground record for the nearest valley station, bias-corrected against the CIMIS reference network, then multiplied by a crop coefficient that tracks the season.
- **The interval comes from the soil, not the calendar.** Farmify tracks how much usable water the root zone still holds and calls a set when the crop is about to feel short — which is a different number of days in April than in July.

- **Each set is sized to what the soil can hold.** Water beyond field capacity is charged as deep percolation rather than quietly ignored, so sets stop being longer than useful.
- **Sets are split when water goes on faster than the ground takes it.** Cycle-and-soak instead of runoff down the row.
- **A water cap becomes a plan rather than a surprise.** Under an allocation, the shortfall is placed on the growth stages that tolerate stress and the sensitive ones are protected, so the season fits the cap instead of running out in August.
- **It corrects itself.** The grower reports whether the ground was too wet, too dry or about right, and the plan for that block moves accordingly.

## 4. Estimated saving

The table below was produced by running Farmify's own engine over six representative valley crop, soil and system combinations, then re-running the identical crop demand, soil reservoir and system efficiency under two common scheduling practices. The only difference between the three rows is *when and how much* water is applied.

| Scheduling practice | Applied water | vs. Farmify |
|---|---|---|
| **Farmify** — demand plus soil water balance | **236 in** | **—** |
| Monthly-adjusted sets with a 15% safety margin<br>the careful grower with no ET data | 278 in | 15% less |
| One set length sized for the peak week, run all season<br>set-and-forget; 27 of 85 inches drained past the roots in the almond case | 409 in | 42% less |

Totals are season-long applied depth summed across the six cases. Farmify produced no crop-stress days in any case, so the reduction is not achieved by under-watering.

### Which figure to believe

Published applied water for mature San Joaquin Valley almonds is about 4.0–4.5 acre-feet per acre. Farmify's own plan for that block came out at 4.19. Average practice is therefore much closer to the monthly-adjusted row than to the set-and-forget row, and **the honest expectation is a 10–20% reduction in applied water**, not 42%. That range also matches the published evidence for ET-based scheduling, which clusters at 10–30% with a central estimate near 20%. The 42% case describes a genuinely badly scheduled block, not the norm.

**For one 100-acre almond block** (419 acre-feet per season under Farmify):

| Reduction | Water saved | Equivalent | Pumping cost saved / year |
|---|---|---|---|
| 10% | 47 acre-feet | 15 million gallons | $7,000 – $14,000 |
| 15% | 74 acre-feet | 24 million gallons | $11,000 – $22,000 |

Cost range assumes $150–300 per acre-foot for pumping and delivery; drought-year market water runs far higher.

**At valley scale**, against roughly 11 million acre-feet of agricultural applied water per year:

| Assumption | Applied water saved | Share of the 1.85 MAF overdraft |
|---|---|---|
| Half the acreage lacks ET scheduling and cuts 10% | 0.55 MAF / year | about 30% |
| Same acreage cuts 20% | 1.10 MAF / year | about 60% |

## 5. The caveat that matters most

Those valley-scale figures are **applied-water** savings, and applied water is not the same as consumed water. This distinction is the single most important qualification in the whole assessment.

### The irrigation efficiency paradox

Water that percolates below the root zone in the San Joaquin Valley largely returns to the same aquifer the grower is pumping from. It is wasted *effort* — lifted, pressurised and paid for — but it is not entirely lost *water*. Worse, the published record shows that on-farm efficiency gains frequently fail to produce basin-scale savings and have sometimes increased total consumption, because growers reinvest the saved water in more acres or thirstier crops.

If 30–50% of the saving is genuinely consumptive loss — surface evaporation, wind drift, canopy wetting, runoff to drains — the real basin saving is nearer **0.25–0.4 MAF per year, or 13–22% of the overdraft**. Material, but not a solution on its own.

What is unambiguously real, regardless of where the water ends up:

- **Energy and cost.** Every acre-foot not pumped is not lifted, not pressurised and not paid for. On a 100-acre block that is five figures a year.
- **Nitrate leaching.** Water draining past the root zone carries fertiliser with it, which is a regulated problem across the valley's east side.
- **Avoided under-watering.** Guesswork errs in both directions. Farmify flags a block that will run short before the crop shows it, which protects yield rather than water.
- **Usable instructions.** The plan is issued in the language spoken on the farm, on one printable page, which is what determines whether any of the above actually happens.

Farmify does not create water. It removes guesswork from a decision that is currently made by calendar and feel on roughly half the valley's acreage, and it makes the resulting plan legible to the person holding the valve.

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**Sources**


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- University of Florida IFAS, *Evapotranspiration-Based Irrigation Scheduling for Agriculture* (AE457); global meta-analysis of irrigation scheduling in apples, peaches and sweet cherries, Agricultural Water Management, 2024.
- USDA ARS and Farm and Ranch Irrigation Survey data on scheduling method adoption; California almond grower irrigation practice survey.
- Allen, Pereira, Raes & Smith, *Crop evapotranspiration*, FAO Irrigation and Drainage Paper 56, 1998 — the basis of Farmify's engine. Doorenbos & Kassam, FAO Paper 33, for yield response.
- California Irrigation Management Information System (CIMIS) reference ET zone normals.


Modelled figures in section 4 were generated by the Farmify engine in this repository (FAO-56 reference ET calibrated to CIMIS, crop coefficients for 18 valley crops, daily soil water balance). They are estimates for comparison between practices, not a prediction for any particular field.